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Question 1

Not answered

Marked out of 4.00

Assessment instruments

For each description below choose an option from the list of scales given here.

A 34-year-old woman complains of loss of weight and appetite for last two months. She has poor sleep and has multiple somatic complaints that started when she and her husband separated three months ago. She denies using alcohol.

A researcher is keen to evaluate the degree of self-rated psychiatric symptoms in a sample of 125 diabetic outpatients attending an insulin clinic.

As a part of a new World Mental Health initiative, it is planned to conduct a nationwide survey of mental health burden. It is intended to administer a fully structured instrument through trained non-clinical research workers to ascertain lifetime diagnosis of mental illness.

A newly constituted mental health trust wants to adopt a standard scale for monitoring clinical recovery on discharge from different adult units.

Positive and negative symptoms scale.
Yale-Brown Obessive scale
General Health Questionnaire
International Personality Disorder Examination
Diagnostic interview scale
Minnesota multiphasic personality inventory
Health of Nations Outcome Scale
CAGE Questionnaire
Global assessment of functioning
Present status examination
Beck's depression scale

Choose...

Your answer is incorrect.

Explanation: Beck's depression scale (BDI) is a self-rated questionnaire containing 21 statements with four possible responses for each.

The General Health Questionnaire (GHQ) is a self-rated screening instrument for the presence of psychiatric illness.

The diagnostic interview schedule (DIS) is a non-clinician administered fully structured interview and used in Epidemiological catchment area study.

The Health of Nations Outcome Scale (HONOS) is used for monitoring clinical recovery and is helpful to measure clinical outcomes in mental health.

The correct answer is: A 34-year-old woman complains of loss of weight and appetite for last two months. She has poor sleep and has multiple somatic complaints that started when she and her husband separated three months ago. She denies using alcohol. → Beck's depression scale, A researcher is keen to evaluate the degree of self-rated psychiatric symptoms in a sample of 125 diabetic outpatients attending an insulin clinic. → General Health Questionnaire, As a part of a new World Mental Health initiative, it is planned to conduct a nationwide survey of mental health burden. It is intended to administer a fully structured instrument through trained non-clinical research workers to ascertain lifetime diagnosis of

mental illness. → Diagnostic interview scale, A newly constituted mental health trust wants to adopt a standard scale for monitoring clinical recovery on discharge from different adult units. → Health of Nations Outcome Scale

Question 2

Not answered

Marked out of 3.00

Pharmacokinetics (1)

Match each characteristic below with the most likely drug from the list given

This drug undergoes no metabolism, and it is excreted unchanged by the kidney

This drug accelerates its own metabolism and the metabolism of other drugs by inducing the hepatic microsomal enzymes

The metabolism of this drug follows zero order kinetics

Nitrazepam
Sodium valproate
Temazepam
Imipramine
Lorazepam
Fluoxetine
Carbamazepine
Phenytoin
Lamotrigine
Diazepam
Lithium

Choose...

Your answer is incorrect.

Lithium is orally well absorbed but not metabolised by the liver; it is renally excreted. 95% of **lithium** is excreted unchanged in the urine.

Carbamazepine is metabolized by the hepatic cytochrome P450 2D6 system, which it also robustly induces. As a result of this autoinduction, the rate of metabolism of carbamazepine (and other P450 substrates) usually increases over the first several weeks of treatment. The initial steady state may be attained within 4 to 5 days, but autoinduction may delay final steady state until 3 to 4 weeks after treatment initiation. Hence, the level of carbamazepine must be monitored, and its dose often must be raised during this early phase of treatment.

Slow release preparations e.g. lithium MR, depot preparations follow zero-order absorption kinetics; drugs that rapidly saturate enzymes such as alcohol and phenytoin follow zero-order elimination kinetics

The correct answer is: This drug undergoes no metabolism, and it is excreted unchanged by the kidney → Lithium, This drug accelerates its own metabolism and the metabolism of other drugs by inducing the hepatic microsomal enzymes → Carbamazepine, The metabolism of this drug follows zero order kinetics → Phenytoin

Question 3

Not answered

Marked out of 3.00

MoK Pharmacology CXEMI017

Drug interactions (1)

Choose agents that will have clinically significant interaction for each description below

1. A 45-year-old patient is an enthusiastic naturalist and is keen on drinking grapefruit juice for breakfast. (Choose TWO) ✖

2. A 35-year-old patient is on long-term NSAID therapy for an ankle injury. (Choose ONE) ✖

- Lithium
- Phenytoin
- Carbamazepine
- Amitryptyline
- St John's wort
- Clomipramine
- Clozapine
- Pimozide

Explanation

The "grapefruit juice effect" is due to components of grapefruit juice (bergamottin, 6, 7-dihydroxybergamottin, and naringenin - these significantly increase drug oral bioavailability by selectively and rapidly down-regulating intestinal (but not liver) CYP3A4 and to a lesser extent, 1A2. This effect is greatest in drugs with high first-pass metabolism such as the dihydropyridine calcium antagonists, felodipine and nimodipine, terfenadine, carbamazepine, triazolam and midazolam (to some extent diazepam), simvastatin and methylprednisone.

Grapefruit juice inhibits the intestinal CYP3A enzymes, resulting in an increase in systemic exposure for CYP3A4 metabolised drugs. Carbamazepine is metabolised primarily by CYP3A4. Pimozide is metabolised by CYP3A and to a lesser extent by CYP1A2. Use of carbamazepine or pimozide with CYP 3A4 inhibitors such as fluoxetine, sertraline, fluvoxamine, nefazodone, and grapefruit juice can raise drug levels. In the case of pimozide, this can increase the risk of dangerous arrhythmias. Buspirone and Pimozide are also significantly affected by grapefruit. (Ref: Psychosomatics; 2005: 46).

The class of medications that could increase serum lithium levels and precipitate lithium toxicity includes diuretics (Loop and Thiazide diuretics), NSAIDs and ACE inhibitors. NSAIDs inhibit the synthesis of renal prostaglandins, thereby reducing renal blood flow and possibly increasing renal re-absorption of sodium and therefore lithium. The magnitude of the rise is unpredictable for any given patient; case reports vary from increases of around 10% to over 400%

Ref: Koziolek M et al. The mechanisms of pharmacokinetic food-drug interactions–A perspective from the UNGAP group. European Journal of Pharmaceutical Sciences. 2019 Jun 15;134:31-59.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 13th Edition. 2018.

The correct answer is:

A 45-year-old patient is an enthusiastic naturalist and is keen on drinking grapefruit juice for breakfast. (Choose TWO) - Carbamazepine, Pimozide

A 35-year-old patient is on long-term NSAID therapy for an ankle injury. (Choose ONE) - Lithium

Question 4

Not answered

Marked out of 4.00

Gait abnormalities in clinical practice
Choose best option for each of the following:

Parkinson's disease	Choose...
Cerebellar disease	Choose...
Upper motor neuron dysfunction	Choose...
Foot drop	Choose...

Pigeon gait
Stiff legged scissoring gate
Failure of gait ignition
Myopic gait
Festinant gait (shuffling gate)
High stepping gait (steppage gait)
Broad-based and unsteady gait
Waddling gait
Lurching chaotic gait
Stomping gait

Your answer is incorrect.

Explanation: In shuffling gait, the individual takes short steps to the point of practically not moving forward. In some patients, the steps vary with a tendency for the patient to accelerate (festinating gait) as he or she walks. This is seen in Parkinson disease.

A patient with ataxia may spread his or her legs apart to widen the base of support to offset the imbalance while standing or walking. The heel-to-toe or tandem walking manoeuvres and standing on one leg uncover subtle forms of ataxia / broad-based gait.

In diplegia (UMN damage) associated with cerebral palsy, the legs are held in adduction while hip and the thighs brush against each other upon walking. A spasm of the inner thigh muscles make the knees tend to slide over each other like the blades of scissors (Scissoring gait).

Steppage gait is seen in chronic peripheral neuropathies and can be the result of the functional elongation of the legs due to bilateral drop foot. The gait is made of high steps as if climbing a flight of stairs while walking on a level surface.

Ref: Neurological History and Physical Examination, http://emedicine.medscape.com/article/1147993-overview (accessed March 31, 2015).

The correct answer is: Parkinson's disease → Festinant gait (shuffling gait), Cerebellar disease → Broad-based and unsteady gait, Upper motor neuron dysfunction → Stiff legged scissoring gait, Foot drop → High stepping gait (Steppage gait)

Question 5

Not answered

Marked out of 5.00

MoK Pharmacology CXEMI023

Antipsychotics & side effects

For each of the following situations, identify the most likely medication associated with it

1. Constipation and paralytic ileus
2. Orthostatic hypotension (choose two answers)
3. Nausea and vomiting
4. Post-injection delirium

Haloperidol
Thioridazine
Trifluoperazine
Olanzapine
Quetiapine
Risperidone
Aripiprazole
Clozapine
Amisulpiride
Sulpiride

Explanation:

Clozapine has pronounced anticholinergic properties. It can cause severe constipation. Clozapine causes constipation by slowing transit time through the gut. Mean transit times are more than four times longer than normal, and 80% of clozapine patients show reduced transit time. Effective treatment or prevention of constipation is essential, as this can prove fatal by progressing to paralytic ileus and bowel ischemia, followed by sepsis and perforation.

Clozapine and Risperidone can cause severe orthostatic hypotension due to alpha one adrenergic blockade.

Nausea, vomiting and gastric side effects are common with Aripiprazole due to 5-HT1A receptor partial agonistic actions.

Olanzapine if administered as long-acting depot injection (Olanzapine Pamoate) could produce symptoms suggestive of antipsychotic overdose in a small number of subjects (< 0.1%), a phenomenon that has been termed "post-injection delirium/sedation syndrome" (PDSS). PDSS is characterized by symptoms related to excessive sedation and delirium soon after the injection, consistent with some of the symptoms of oral olanzapine overdose.

Ref: McDonnell et al., BMC Psychiatry. 2010; 10: 45. Retrieved from http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2895590/

The correct answer is:

Constipation & Paralytic Ileus - Clozapine,

Orthostatic Hypotension - Clozapine,Risperidone

Post-injection delirium - Olanzapine,

Nause and Vomiting - Aripiprazole

Question 6

Not answered

Marked out of 5.00

Disturbed thinking processes
For each description given below choose the most appropriate term from the list above;

Replies to questions are irrelevant or oblique. The response usually refers to the relevant topic but fails to give a complete answer

The individual cannot express ideas as quickly as they come into his head, presenting with fragmented thoughts, abrupt changes in the topic and general incoherence.

When answering questions, the patient gets stuck on the last syllable of a word and repeats it

Abrupt interruption in a train of thinking before an idea is complete

Rapid production of speech output with a subjective feeling of racing thoughts

Circumstantiality
Flight of ideas
Clanging
Perseveration
Logoclonia
Derailment
Thought blocking
Pressure of speech
Neologisms
Tangentulay

Choose...

Your answer is incorrect.

Explanation: In tangentiality the replies to questions are irrelevant or oblique. The response usually refers to the appropriate topic but fails to give a complete answer.Tangentiality is an oblique, digressive, or even irrelevant manner of speech in which the central idea is not communicated. It is characterised by producing answers which are only indirectly related to the question asked by the examiner.

Circumstantiality occurs when thinking proceeds slowly with many unnecessary and trivial details, but finally the point is reached.

The patient never reaches the point in tangentiality, whereas they do reach the point in circumstantiality.

In flight of ideas, the individual cannot express ideas as quickly as they come into his head, leading to fragmented thoughts, abrupt changes in the topic and general incoherence. It is characteristic of mania.

In logoclonia, the patient gets stuck on the last syllable of a word and repeats it. It is often a symptom of Parkinson's disease.

Thought blocking (or deprivation) is the abrupt interruption in a train of thinking before a thought or idea is finished. Following a pause, the patient may report no recall of what was being said or what was going to be said.

Pressure of speech refers to the rapid production of speech output with a subjective feeling of racing thoughts often seen in mania.

The correct answer is: Replies to questions are irrelevant or oblique. The response usually refers to the relevant topic but fails to give a complete answer → Tangentiality, The individual cannot express ideas as quickly as they come into his head, presenting with fragmented thoughts, abrupt

changes in the topic and general incoherence. → Flight of Ideas, When answering questions, the patient gets stuck on the last syllable of a word and repeats it → Logoclonia, Abrupt interruption in a train of thinking before an idea is complete → Thought blocking, Rapid production of speech output with a subjective feeling of racing thoughts → Pressure of speech

Question 7

Not answered

Marked out of 3.00

Sensory disturbances

For each of the description below choose one correct option from the given list:

A 22-year-old man can smell music after consuming an unknown drug cocktail

A 44-year-old man walking down a dimly lit road perceives fearsome shadows beneath trees

A 77-year-old man admitted with delirium has altered sensory threshold and the nurses have to shout aloud to communicate with him

Hypoacusis
Somatic passvity
Delusional misperception
Hyperesthesia
Illusions
Delusional misrepresentation
Synaesthesia
Reflex hallucination
Allodynia
Delusional perception
Referential delusion

Your answer is incorrect.

Explanation: Synaesthesia is characterised by object perception in a modality different from that of the original stimulus. It is the experience of a stimulus in one sense modality producing a sensory experience in another (e.g. 'tasting' sounds or 'hearing' colours). It can occur in hallucinogenic drug intoxication and in epileptic states.

An illusion is a type of false perception in which the perception of a real-world object is combined with internal imagery to produce a false internal percept. Illusion described in question 2 refers to affect illusion where a fearful emotional state contributes to misperceptions.

In some cases of altered sensorium (delirium), hypoacusis is commonly seen.

Hypoacusis occurs in delirium, where the threshold for all sensations is raised. The defect of attention found in delirium further reduces sensory acuity. This highlights the importance of speaking to delirious patients more loudly and more slowly than usual.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019.

The correct answer is: A 22-year-old man can smell music after consuming an unknown drug cocktail → Synaesthesia, A 44-year-old man walking down a dimly lit road perceives fearsome shadows beneath trees → Illusions, A 77-year-old man admitted with delirium has altered sensory threshold and the nurses have to shout aloud to communicate with him → Hypoacusis

Question 8

Not answered

Marked out of 3.00

Jung and Analytical Psychology

For each description given below, select the most appropriate Jungian concept from the list above

A concept regarded as universally common to mankind. It contains various archetypes described by Jung.

Images and symbols that represent human experience common to different cultures

The unconscious feminine aspects of man

Shadow
Collective unconscious
Archetypes
Extroversion
Personal unconscious
Persona
Introversion
Anima
Animus
Analytical perception
Self

Your answer is incorrect.

Explanation: Jung founded analytic psychology and introduced the terms extraversion and introversion. He described collective unconscious - a term to describe all of the mankind's collective symbolic past, and archetypes - like images and symbols that constitute the collective unconscious. These representational images have universal symbolic meaning (e.g. Hero, Old Wise Man, Tree, etc.).

Other terms described by Jung include Persona which is the mask covering one's personality and presented to outside world, Anima - unconscious feminine aspect of a man, Animus - unconscious masculine character of a woman, Shadow - an archetype - a personification of unacceptable aspects of oneself symbolized as a dark internal alien and Individuation - ultimate goal of life where an individual develops a sense of self- identity.

The correct answer is: A concept regarded as universally common to mankind. It contains various archetypes described by Jung. → Collective unconscious, Images and symbols that represent human experience common to different cultures → Archetypes, The unconscious feminine aspects of man → Anima

Question 9

Not answered

Marked out of 3.00

Blood supply to the brain

Match the specific areas of the brain with their blood supply from the list provided:

Most of cerebellar hemisphere	Choose...
Posterior limb of internal capsule	Choose...
Cerebellum-vermis	Choose...

Medial cerebral artery
Posterior inferior cerebellar artery
Subclavian artery
Anterior communicating artery
Anterior cerebral artery
Internal auditory artery
Superior cerebellar artery
Anterior inferior cerebellar artery
Posterior cerebral artery

Your answer is incorrect.

Most of the cerebellar hemispheres including superior and middle cerebellar peduncles and cerebellar nuclei are supplied by the superior cerebellar artery. The vermis of the cerebellum is supplied by the posterior inferior cerebellar artery.

The posterior limb of the internal capsule is supplied by the posterior cerebral artery.

The correct answer is: Most of cerebellar hemisphere → Superior cerebellar artery, Posterior limb of internal capsule → Posterior cerebral artery, Cerebellum-vermis → Posterior inferior cerebellar artery

Question 10

Not answered

Marked out of 3.00

Neurological symptoms

Find one term each to describe the following presentations:

A 65-year-old gentleman was unable to identify familiar objects such as key or coins by touch with eyes closed. He has difficulty in describing their shape and texture. However, he identifies them when eyes are open.

A 67-year-old woman was unable to recognize what number is scratched into her skin without seeing

A man has left side hemiplegia after a stroke. He is unaware of his deficit.

Constructional apraxia
Dressing apraxia
Agraphaesthesia
Prosopagnosia
Finger agnosia
Visuospatial agnosia
Anosognosia
Right-left disorentation
Astereognosis

Your answer is incorrect.

Agraphaesthesia is defined as inability to recognize what number or alphabet is scratched on one's skin without seeing.

Astereognosis is defined as inability to recognize objects by palpation (touch), and without visual inspection.

Anosognosia refers to the inability to identify one's neurological deficits.

The correct answer is: A 65-year-old gentleman was unable to identify familiar objects such as key or coins by touch with eyes closed. He has difficulty in describing their shape and texture. However, he identifies them when eyes are open. → Astereognosis, A 67-year-old woman was unable to recognize what number is scratched into her skin without seeing → Agraphaesthesia, A man has left side hemiplegia after a stroke. He is unaware of his deficit. → Anosognosia

Question 11

Not answered

Marked out of 5.00

Receptor localization

For each of these areas of the brain, identify the main neurotransmitter pathways localized to the specified regions:

Median raphe nuclei	Choose...
Nucleus basalis of Meynert	Choose...
Substantia nigra	Choose...
Dorsal raphe nuclei	Choose...
Locus coeruleus	Choose...

Glutamate
Serotonin
Dopamine
Opiate receptors
Norepinephrine
GABA
Glycine
Acetylcholine
Histamine

Your answer is incorrect.

The major site of serotonergic cell bodies is in the upper pons and midbrain - in the median and dorsal raphe nuclei. Serotonin has both ascending and descending projections. Ascending serotonergic projections originate mainly in the raphe nucleus in the brainstem. These ascending projections may regulate mood, anxiety, sleep, and other functions. Descending serotonergic projections extend down the brainstem and through the spinal cord. They may regulate pain.

The nucleus basalis of Meynert is a large source of cholinergic innervation to widespread cortical areas. Several studies have reported data consistent with the hypothesis that specific degeneration of cholinergic neurons is present in the nucleus basalis of Meynert in people with Alzheimer's disease. A group of acetylcholine producing neurons in the nucleus basalis of Meynert project to the cerebral cortex, limbic system, hypothalamus and thalamus.

Norepinephrine-producing neurons are found within the brain in the pons and medulla in two major clusterings: the locus coeruleus and the lateral tegmental noradrenergic nuclei.The major concentration of norepinephrine is in the cell bodies that project upward in the brain is in the locus coeruleus in the pons.

A key dopamine pathway is the nigrostriatal pathway, which projects from dopamine cell bodies in the brainstem substantia nigra via axons terminating in the striatum. Dopamine is mainly localized in nigrostriatal pathways, mesolimbic and mesocortical pathways and tuberoinfundibular pathways.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022.

The correct answer is: Median raphe nuclei → Serotonin, Nucleus basalis of Meynert → Acetylcholine, Substantia nigra → Dopamine, Dorsal raphe nuclei → Serotonin, Locus coeruleus → Norepinephrine

Question 12

Not answered

Marked out of 5.00

Lobar dysfunction

For each of the clinical descriptions given below, identify the area of the brain damaged.

Alexia without agraphia, colour agnosia and visual object agnosia	Choose...
Profound changes in personality, problems with verbal expression and forced utilization behaviour	Choose...
Profound amnesia	Choose...
Visuospatial agnosia, prospagnosia and complex visual hallucinations	Choose...
Dressing apraxia, prospagnosia, hemisomatognosia	Choose...

Dominant temporal lobe
Dominant parietal lobe
Frontal lobe
Non-dominant temporal lobe
Non-dominant occipital lobe
Cerebellum
Non-dominant parietal lobe
Bilateral temporal lobe lesions
Dominant occipital lobe

Your answer is incorrect.

Lesions in the dominant occipital lobe result in alexia without agraphia, colour agnosia and visual object agnosia.

Lesions in the non-dominant occipital lobe result in visuospatial agnosia, metamorphopsia (image distortion) and complex visual hallucinations.

Damage to the frontal lobes results in a change in one's personality, with poor motivation, poor judgement and impaired social awareness. If the orbital frontal region (just above the eye sockets) is damaged, forced utilization behaviour may be seen. Here, when objects are placed in front of the subject, they will be picked up and used appropriately even when the subject is told not to do so.

Profound amnesia results from bilateral medial temporal lobe lesions. The causes of such lesions would include infections, tumour and epilepsy. Both anterograde and retrograde memory will be impaired with intact immediate memory (i.e. unimpaired digit span) and preserved executive task performance.

Lesions in the non-dominant parietal lobe result in anosognosia, hemisomatagnosia, dressing apraxia and prosopagnosia. Gerstmann syndrome is associated with lesions of the dominant parietal lobe.

(Ref: Core Psychiatry: pg 16-20)

The correct answer is: Alexia without agraphia, colour agnosia and visual object agnosia → Dominant occipital lobe, Profound changes in personality, problems with verbal expression and forced utilization behaviour → Frontal lobe, Profound amnesia → Bilateral temporal lobe lesions, Visuospatial agnosia, prospagnosia and complex visual hallucinations → Non-dominant occipital lobe, Dressing apraxia, prospagnosia, hemisomatognosia → Non-dominant parietal lobe

Question 13

Not answered

Marked out of 3.00

Pharmacodynamics

Select one mechanism of action each for the following drugs

Rivastigmine	Choose...
Selegeline	Choose...
Diazepam	Choose...

GABA-B agonist
MAO-B inhibitor
5HT1A partial agonist
D-2 receptor partial agnoist
MAO-A inhibitor
Presynaptic alpha-2 agonist
Butyryl-cholinesterase inhibitor
Selective norepinephrine and serotonin re-uptake
NMDA receptor antagonist

Your answer is incorrect.

Rivastigmine is a cholinesterase Inhibitor. It acts by inhibiting acetylcholinesterase enzyme that breaks down acetylcholine centrally. Rivastigmine inhibits both the acetyl and butyl-cholinesterase while donepezil and galantamine are acetyl specific.

Selegiline is a monoamine oxidase Inhibitor - selective for the enzyme MAO-B at normal therapeutic doses; selectivity is lost when the drug is administered (usually as a patch) at higher doses, leading to some antidepressant action.

Benzodiazepines belong to a class of drugs that modulate the neurotransmitter GABA, especially GABA-A, to open the chloride channel. Benzodiazepines have a powerfully calming effect on the brain. They can modulate GABA-A receptors as an agonist. GABA has a fantastic property of being an inhibitory neurotransmitter with the ability to reduce the excitability of neurons. Diazepam is a type of benzodiazepine which acts via a specific site called omega site in GABA-A complex. All benzodiazepines are full agonists except clonazepam which is a partial agonist. They facilitate GABA action on GABA-A complex - thus inhibitory neurotransmission via chloride ion flow is facilitated. Benzodiazepines have no direct agonistic action in the absence of GABA.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021.

Fluyau D et al. Challenges of the pharmacological management of benzodiazepine withdrawal, dependence, and discontinuation. Therapeutic advances in psychopharmacology. 2018 May;8(5):147-68.

The correct answer is: Rivastigmine → Butyryl-cholinesterase inhibitor, Selegeline → MAO-B Inhibitor, Diazepam → GABA-A agonist

Question 14

Not answered

Marked out of 1.00

Which of the following brain regions is supplied by the anterior cerebral artery?

Select one:

- cuneus
- Inferior temporal gyrus
- Lingual gyrus
- Genu of corpus callosum
- planum temporale

Your answer is incorrect.

MCA supplies blood to the cortical areas involved in speech, swallowing and language, including the lateral motor strip, lateral sensory strip, Broca's area, Wernicke's area, Heschl's gyrus, planum temporale and the angular gyrus. In addition, it provides most of the blood supply to the corpus striatum. PCA supplies the cuneus, inferior temporal gyrus and lingual gyrus.

Genu and 3/4ths of corpus callosum are supplied by ACA. The anterior cerebral artery territory consists of the superior frontal gyrus, inferior and medical surfaces of the frontal lobe to the level of the precuneus, genu and body of the corpus callosum, and anterior portion of the basal ganglia supplied by the recurrent artery of Heubner.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019.

The correct answer is: Genu of corpus callosum

Question 15

Not answered

Marked out of 1.00

Which region of cerebral blood flow correlates with hallucinations and delusions?

Select one:

- Right medial temporal cortex
- Left medial temporal cortex
- Caudate nucleus
- Left prefrontal cortex
- Right prefrontal cortex

Your answer is incorrect.

Patients with schizophrenia who experience frequent auditory hallucinations show reduced grey matter density and neuronal activation. Similarly, f-MRI images have shown reduced activation in the left temporal lobe and anterior cingulate cortex. The brain structural and functional abnormalities seem to converge in the superior gyrus of the left temporal lobe.

The correct answer is: Left medial temporal cortex

Question 16

Not answered

Marked out of 1.00

The role of ependymal cells is

Select one:

- Lining the walls of the ventricular systems
- Nutrition of neurons
- Produce myelin sheaths
- Clear neuronal debris
- Maintains the blood brain barrier

Your answer is incorrect.

Ependymal cells are any of the neuroglial cells of the ependyma in the ventricles of the brain, with cilia on their surfaces helping to propel cerebrospinal fluid through the ventricular system. Specialised ependymal cells form the choroid plexus.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Lining the walls of the ventricular systems

Question 17

Not answered

Marked out of 1.00

The disengagement of attention is mediated by

Select one:

- Medial temporal cortex
- Medial parietal cortex
- Lateral temporal cortex
- Prefrontal cortex
- Occipital cortex

Your answer is incorrect.

The parietal lobe is postulated to be involved in engagement and disengagement of attention. In conditions like Alzheimer's disease, the medial parietal cortex is affected in early stages, and attentional deficits are seen before language and visuospatial deficits.

The correct answer is: Medial parietal cortex

Question 18

Not answered

Marked out of 1.00

Which neuroglial cells act as scavenger cells at the sites of CNS injury?

Select one:

- Schwann cells
- Microglia
- Satellite cells
- Astrocytes
- Ependymal cells

Your answer is incorrect.

Microglia acts as scavenger cells at sites of CNS injury. **Microglia,** which are derived from macrophages, are involved in removing cellular debris following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022.

Sasaki A. Microglia and brain macrophages: an update. Neuropathology. 2017 Oct;37(5):452-64

The correct answer is: Microglia

Question 19

Not answered

Marked out of 1.00

Which of the following enzymes is seen only in the neuron and not in the glial cells?

Select one:

- Glutamic Acid Decarboxylase
- monoamine oxidase
- dopadecarboxylase
- cholinesterase
- Glutamine Synthase

Your answer is incorrect.

Glutamic acid decarboxylase [GAD] is a neuron-specific enzyme that is not found in glial cells. Hence, antibodies to GAD can be used to identify neurons that release GABA. Glutamine synthase is a glial enzyme that metabolizes glutamate into glutamine.

The correct answer is: Glutamic Acid Decarboxylase

Question 20

Not answered

Marked out of 1.00

MAO-A is available primarily in which of the following body tissues?

Select one:

- Spleen
- Lungs
- Brain
- Kidney
- Gut

Your answer is incorrect.

Monoamine oxidases are of two types: the brain predominantly has monoamine oxidase inhibitor -B and in the liver/GI tract, predominantly monoamine oxidase inhibitor-A is noted.

Historically, most studies on the function and localisation of MAO enzymes have focused on the distribution and role of MAOs in the central nervous system, although widespread expression of MAO mRNA is found. MAO-A is preferentially expressed in the gastrointestinal tract and found in moderately high levels in human heart.

Ref: Ostadkarampour M, Putnins EE. Monoamine Oxidase Inhibitors: A Review of Their Anti-Inflammatory Therapeutic Potential and Mechanisms of Action. Frontiers in Pharmacology. 2021 Apr 30;12:889.

The correct answer is: Gut

Question 21

Not answered

Marked out of 1.00

The most common side effect when using sildenafil is:

Select one:

- Headache
- Insomnia
- Double vision
- Hypotension
- Bradycardia

Your answer is incorrect.

Viagra is an indirect vasodilator, resulting in headaches.

Common side effects of phosphodiesterase inhibitors (such as sildenafil and tadalafil) are mild headaches, dizziness, and nasal congestion. (Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 13th Edition. 2018).

The correct answer is: Headache

Question 22

Not answered

Marked out of 1.00

Low CSF concentration of 5-Hydroxy Indole Acetic Acid is associated with which of the following?

Select one:

- Cheese Reaction
- Serotonin Syndrome
- Impulsivity
- Visual Hallucinations
- Coronary Artery Disease

Your answer is incorrect.

Animal and human research suggests that the central serotonin system is involved in the inhibition of impulsive behaviour A low CSF 5-HIAA concentration was primarily associated with impulsivity. (http://archpsyc.ama-assn.org/cgi/content/abstract/51/1/20).

Much of the clinical data relating 5-HT and impulsivity have been provided by investigations of suicide. Early studies reported lower CSF and plasma 5-HIAA levels as well as blunted prolactin response to fenfluramine in both suicide attempters and completers, as well as lower brain 5-HT, 5-HT transporter, and abnormal 5-HT receptor binding at post-mortem in suicide completers. Research into individuals with antisocial personality disorder categorised according to whether their violent behaviour was aggressive or non-aggressive has revealed similar results, with impulsive aggressive individuals reported to have lower levels of CSF 5-HIAA and blunted prolactin response to fenfluramine.

Dalley JW, Roiser JP. Dopamine, serotonin and impulsivity. Neuroscience. 2012 Jul 26;215:42-58.

The correct answer is: Impulsivity

Question 23

Not answered

Marked out of 1.00

Nitric oxide is synthesized from which of the following?

Select one:

- Toluene
- Benzene
- Phosphokinase
- L-arginine
- L-tryptophan

Your answer is incorrect.

Nitric oxide is synthesized from L-arginine when stimulated by nitric oxide synthase.

L-arginine is the substrate for nitric oxide synthases, which then produces nitric oxide and L-citrulline.

Ref: Cziráki A et al. L-Arginine-nitric oxide-asymmetric dimethylarginine pathway and the coronary circulation: translation of basic science results to clinical practice. Frontiers in Pharmacology. 2020;11.

The correct answer is: L-arginine

Question 24

Not answered

Marked out of 1.00

Which of the following human spongiform encephalopathy has the slowest course?

Select one:

- Sporadic CJD
- Variant CJD
- Iatrogenic CJD
- Gerstmann- Straussler- scheinker syndrome
- Kuru

Your answer is incorrect.

Kuru is a type of prion disease which has the slowest course. The incubation period was prolonged- up to 40 years before disease onset, which was then rapid and fatal. It is thought to have been transmitted by ritual cannibalism. Symptoms include personality and behavioural disturbances, progressive cerebellar ataxia, dementia and death.

The correct answer is: Kuru

Question 25

Not answered

Marked out of 1.00

Mr. X is a 73-year-old gentleman with 5-year history of Parkinson's disease, cognitive decline and visual hallucinations. Post mortem evidence suggests alpha syn-nuclein deposits and cytoplasmic inclusions. The common microscopic finding expected in this case would be

Select one:

- Pick cells
- Tau proteins
- Hirano bodies
- Lewy bodies
- Pick bodies

Your answer is incorrect.

The clinical features, in this case, are suggestive of Parkinson's disease dementia. In Parkinson's disease, the Lewy bodies are confined to the substantia nigra. The Lewy bodies are weak eosinophilic, spherical, cytoplasmic inclusions. Parkinson's disease and Lewy body dementia contains an accumulation of alpha-synuclein and is classified under synucleinopathies.

The key constituent of Lewy bodies is alpha-synuclein. The function of alpha-synuclein is not certain, but it is through to function in cell membrane remodeling at neuronal terminals. When aggregates of misfolded, overexpressed alpha-synuclein accumulate in neurons or surrounding glial cells, Lewy bodies are formed.

Ref: Sanford AM. Lewy body dementia. Clinics in geriatric medicine. 2018 Nov 1;34(4):603-15.

The correct answer is: Lewy bodies

Question 26

Not answered

Marked out of 1.00

Which of the following neuroimaging finding is most suggestive of Alzheimer's disease?

Select one:

- Occipital hypoperfusion on HMPAO- SPECT
- Impaired dorsolateral cortex function
- Activation of primary auditory cortex on functional MRI
- Increased signal in the Pulvinar nucleus of thalamus bilaterally
- Medial temporal atrophy on structural MRI scan

Your answer is incorrect.

MRI in Alzheimer's disease often shows atrophy of grey matter involving the hippocampus, entorhinal cortex and amygdala in the medial temporal lobe.

The entorhinal cortex, an essential component of the medial temporal lobe long-term-memory system, represents the main source of input to the hippocampus and the primary target of hippocampal outputs. The involvement of the entorhinal cortex is relevant for neurodegenerative disorders such as Alzheimer's disease (AD), as it is one of the earliest affected brain regions. The typical hallmarks of AD, such as the presence of amyloid protein and neurofibrillary tangles, are seen primarily in the entorhinal cortex in mild AD and 'spread' to the hippocampus and other cortical areas as the disease progresses.

Macroscopically, the AD brain often has at least moderate cortical atrophy that is most marked in multimodal association cortices and limbic lobe structures. The frontal and temporal cortices often have enlarged sulcal spaces with atrophy of the gyri. There is also increasing recognition of atrophy in posterior cortical areas in AD. As a result of this atrophy, there is often enlargement of the frontal and temporal horns of the lateral ventricles, and decreased brain weight is observed in most affected individuals.

Ref: Criscuolo C et al. Entorhinal Cortex dysfunction can be rescued by inhibition of microglial RAGE in an Alzheimer's disease mouse model. Scientific reports. 2017 Feb 13;7(1):1-5.

DeTure MA, Dickson DW. The neuropathological diagnosis of Alzheimer's disease. Molecular neurodegeneration. 2019 Dec;14(1):1-8.

The correct answer is: Medial temporal atrophy on structural MRI scan

Question 27

Not answered

Marked out of 1.00

Which of the following neural structures is strongly implicated in the regulation of P50 suppression, a proposed endophenotype in schizophrenia?

Select one:

- Frontal cortex
- Angular gyrus
- Basal ganglia
- Hippocampus
- Ventral tegmentum

Your answer is incorrect.

P50 suppression is regulated by wide-ranging neural circuitry, prominently involving hippocampal structures.

The ability to successfully filter or gate sensory information is an important feature of a healthy individual. Patients with schizophrenia show deficits in sensory gating which theoretically may contribute to a state of flooding of higher brain functions and ultimately may lead to the formation of psychoses. Early aspects of sensory gating can be quantified with a paired stimulus paradigm, in which 2 identical clicks of sound are presented with a 500 ms interstimulus interval. A healthy subjects' P50 amplitude is usually significantly decreased in response to the second of these clicks. This decreased response is assumed to be the result of the first stimulus being processed by the brain, which effectively gates the processing of the second stimulus. There are many studies showing that P50 suppression is reduced in patients with schizophrenia. The sources of P50 suppression are largely unknown, but a hippocampal involvement has frequently been proposed.

Ref: Oranje B et al. P50 suppression and its neural generators in antipsychotic-naive first-episode schizophrenia before and after 6 months of quetiapine treatment. Schizophrenia bulletin. 2013 Mar 1;39(2):472-80.

The correct answer is: Hippocampus

Question 28

Not answered

Marked out of 1.00

A 56-year-old lady was admitted with depression following a suicide attempt by self-induced skin laceration. She was increasingly isolating herself and complained of somatic pain symptoms in her back. She also had constipation for few weeks. No past psychiatric history was noted. Routine blood tests revealed an isolated anomaly of high serum calcium level (15.8 mg/dL). Which of the following is the most likely diagnosis?

Select one:

- Chronic renal failure
- Hyperparathyroidism
- Hypothyroidism
- Hyperprolactinemia
- Hypoparathyroidism

Your answer is incorrect.

Primary hyperparathyroidism is generally related to parathyroid adenomas. Most patients are asymptomatic but can have recurrent kidney stones, peptic ulcers, dehydration, constipation, and altered mental status. Diagnosis can be confirmed with measurement of parathyroid hormone levels. (Ref: http://psy.psychiatryonline.org/cgi/content/full/40/6/486).

Hyperparathyroidism has traditionally been recognised by its characteristic symptoms, including urolithiasis ('stones'); osteopenia and osteoporosis ('bones'); abdominal cramping, nausea, and peptic ulceration ('moans'); and depression, anxiety, cognitive dysfunction, insomnia, confusion, and personality changes ('psychiatric overtones'). Calcium is thought to figure prominently in determining changes in monoamine metabolism in the CNS, thereby modifying neurotransmission and resulting in alterations in mood and cognition. (Ref: Parks KA et al. Psychiatric complications of primary hyperparathyroidism and mild hypercalcemia. American Journal of Psychiatry. 2017 Jul 1;174(7):620-2).

The correct answer is: Hyperparathyroidism

Question 29

Not answered

Marked out of 1.00

Which of the following statements with regard to alpha waves in the EEG is correct?

Select one:

- Antipsychotics abolish alpha waves
- It is maximal over occipital region
- It reduces with eye closure
- Excess of alpha rhythm is seen in Huntington's disease
- It is a higher frequency band than beta waves

Your answer is incorrect.

The alpha brain waves oscillate about 10 times per second, and the range is 8-13 cycles per second. The brain waves called 'Alpha' were the first to be discovered (around 1908, by an Austrian Psychiatrist named Hans Berger).Neuroleptics commonly produce increased alpha rhythm without obvious clinical manifestations. In Huntington's the EEG will be low voltage, with poorly formed or absent alpha waves.

The correct answer is: It is maximal over occipital region

Question 30

Not answered

Marked out of 1.00

With respect to depression seen in patients with schizophrenia, which of the following is correct?

Select one:

- It is seen only in post-psychotic period
- It is not an outcome indicator
- It often coexists with negative symptoms
- It is often a side effect of antipsychotic use
- It does not respond to antidepressants

Your answer is incorrect.

Depression can be seen in either acute or postpsychotic phases of schizophrenia. Post-schizophrenic depression is diagnosed if the general criteria for schizophrenia were met within the past twelve months, but not in full at the present time. But one of either positive/negative symptoms/formal thought disorder/catatonia symptoms must be present.

Affective symptoms have been recognised as occurring commonly in non-affective psychosis, in 50-80% of patients at some point during their illness. Up to half of individuals with post-psychotic depression may also meet criteria for negative symptoms, implying a syndromal overlap between depressive and negative symptoms.

Ref: Krynicki CR et al. The relationship between negative symptoms and depression in schizophrenia: a systematic review. Acta Psychiatrica Scandinavica. 2018 May;137(5):380-90.

The correct answer is: It often coexists with negative symptoms

Question 31

Not answered

Marked out of 1.00

Which of the following is true with regard to posttraumatic stress disorder? (according to ICD-10 criteria)

Select one:

- Diagnostic EEG findings often noted
- Not seen in children
- Patients show decreased startle reaction
- Onset is within 6 months of trauma
- Trauma must be near fatal in nature

Your answer is incorrect.

The onset of PTSD is generally within six months of the trauma. Retrospective reports of war veterans reveal that delayed-onset PTSD is extremely rare one year post-trauma, and there was no evidence of PTSD symptom onset 6 or more years after trauma exposure. Br J Psychiatry. 2009 Jun;194(6):515-20.

The correct answer is: Onset is within 6 months of trauma

Question 32

Not answered

Marked out of 1.00

Select one typical feature of hypomania

Select one:

- Presence of bizarre delusions
- Presence of mood incongruent delusions
- Absence of flight of ideas
- Absence of significant psychosocial dysfunction
- Presence of hallucinations

Your answer is incorrect.

As defined in ICD-11, a hypomanic episode is a persistent mood state lasting at least several days characterised by mild elevation of mood or increased irritability and increased activity of a subjective experience of increased energy, accompanied by other characteristic symptoms such as rapid speech, rapid or racing thoughts, increased self-esteem, an increase in sexual drive or sociability, decreased need for sleep, distractibility, or impulsive or reckless behaviour. The symptoms are not severe enough to cause marked impairment in occupational functioning or in usual activities or relationships with other, does not necessitate hospitalisation, and there are no accompanying delusions or hallucinations.

The correct answer is: Absence of significant psychosocial dysfunction

Question 33

Not answered

Marked out of 1.00

What is the most common psychiatric disorder in children with mild intellectual disability?

Select one:

- Conduct disorder
- Psychosis
- Autistic spectrum disorder
- Personality disorder
- Depression

Your answer is incorrect.

Disruptive and conduct-disorder behaviours occurred more commonly in the mildly disabled group, which represents approximately 85 percent of persons with intellectual disability. ADHD is often a prominent feature in children with ID. Those with severe mental retardation have a particularly high rate of autistic disorder and pervasive developmental disorders.

The correct answer is: Conduct disorder

Question 34

Not answered

Marked out of 1.00

A 45-year-old woman believes that her neighbours are plotting to kill her for the last 6 months. There is no hallucination. The most likely diagnosis is

Select one:

- Adjustment disorder
- Dementia
- Schizophrenia
- Bipolar disorder
- Delusional disorder

Your answer is incorrect.

Delusional disorder is an illness characterized by the presence of nonbizarre delusions in the absence of other mood or psychotic symptoms.

Delusional disorder is characterised by the development of a delusion or set of related delusions, typically persisting for at least 3 months and often much longer, in the absence of a depressive, manic, or mixed mood episode. The delusions are variable in content across individuals, but typically stable within individuals, although they may evolve over time. Other characteristic symptoms of schizophrenia (i.e. clear and persistent hallucinations, negative symptoms, disorganised thinking, or experiences of influence, passivity, or control) are not present, although various forms of perceptual disturbances (e.g. hallucinations, illusions, misidentification of persons) thematically related to the delusion are still consistent with the diagnosis.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed,; ICD-11; World Health Organization, 2020)

The correct answer is: Delusional disorder

Question 35

Not answered

Marked out of 1.00

Mrs. Chomes has recently lost her husband after his protracted fight with stomach cancer. Which of the following is indicative of a pathological grief disorder?

Select one:

- Signing up for counselling
- Persistence of grief response for more than 12 months after his death
- Joining a charity for fighting prostate cancer
- Clearing his room of his belongings
- Visual hallucinations of her husband

Your answer is incorrect.

Prolonged grief disorder is a disturbance in which, following the death of a partner, parent, child, or other person close to the bereaved, there is persistent and pervasive grief response characterised by longing for the deceased or persistent preoccupation with the deceased accompanied by intense emotional pain (e.g. sadness, guilt, anger, denial, blame, difficulty accepting the death, feeling one has lost a part of one's self, an inability to experience positive mood, emotional numbness, difficulty in engaging with social or other activities).

The grief response has persisted for an atypically long period of time following the loss (more than 6 months at a minimum) and clearly exceeds expected social, cultural, or religious norms for the individual's culture and context. Grief reactions that have persisted for longer periods that are within a normative period of grieving given the person's cultural and religious context are viewed as normal bereavement responses and are not assigned a diagnosis. The disturbance causes significant impairment in personal, family, social, educational, occupational. or other important areas of functioning.

Some degree of mummification (preserving the belongings of the deceased as they were before death), visual hallucinations of the deceased ('sighting the dead'), altruistic charity and seeking help through counseling are normal reactions.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed,; ICD-11; World Health Organization, 2020)

The correct answer is: Persistence of grief response for more than 12 months after his death

Question 36

Not answered

Marked out of 1.00

In ICD-10 astasia-abasia is described under:

Select one:

- Dissociative fugue disorder
- Dissociative motor disorder
- Factitious disorder
- Malingering
- Dissociative stupor

Your answer is incorrect.

In ICD10 dissociative (conversion) disorders are classified into dissociative amnesia, dissociative fugue, trance /possession, disorders of movement (includes motor disorders), dissociative stupor, dissociative convulsions (pseudoseizures) and dissociative sensory loss / anaesthesia. Astasia-abasia is discussed under dissociative motor disorders.

The correct answer is: Dissociative motor disorder

Question 37

Not answered

Marked out of 1.00

A university student has been diagnosed with Kleine Levin syndrome after presenting with recurring periods of excessive drowsiness and sleep. Which of the following feature is NOT a part of this diagnosis?

Select one:

- Irritability
- Excessive food intake
- Drop attacks
- Sexual disinhibition
- Confusion

Your answer is incorrect.

Drop attacks are not a part of Kleine Levin syndrome.

Kleine-Levin syndrome (KLS) is a rare disorder characterised by periodic episodes of hypersomnolence and hyperphagia. It is characterised by periods lasting from days to weeks of attacks of hypersomnia, accompanied by excessive food intake. Other behavioural symptoms may occur, including sexual disinhibition (which may appear compulsive in nature), along with a variety of other psychiatric symptoms such as confusion, irritability, restlessness, euphoria, hallucinations, delusions, and schizophreniform states.

Laboratory tests may show slight changes in the electroencephalogram. However, clinical presentation and laboratory tests are normal during asymptomatic intervals. KLS most often presents in adolescent males, with complete recovery by the 3rd to the 4th decade of life. Possible precipitating factors include excessive workload, febrile illness, and respiratory infections (Kleine-Levin syndrome: a unique cause of fatigue. Retrieved from http://bjsm.bmj.com/content/39/2/e7.full.html).

The correct answer is: Drop attacks

Question 38

Not answered

Marked out of 1.00

Which of the following disorders is included in the new diagnostic grouping of disorders of bodily distress and bodily experience in ICD-11?

Select one:

- Body dysmorphic disorder
- Excoriation disorder
- Hypochondriasis
- Body integrity dysphoria
- Trichotillomania

Your answer is incorrect.

The diagnostic grouping of disorders of bodily distress and bodily experience includes bodily distress disorder, which broadly replaces somatoform disorders and neurasthenia, and body integrity dysphoria, which is characterised by the desire to have a certain disability. The other diagnoses lie within the obsessive-compulsive and related disorders, with trichotillomania and excoriation disorder being within the body-focused repetitive behaviour disorders subgrouping.

The correct answer is: Body integrity dysphoria

Question 39

Not answered

Marked out of 1.00

Mr Richmond obtains gratification by rubbing against women travelling on the crowded Dockland Railways. What is this condition called?

Select one:

- Exhibitionistic disorder
- Voyeuristic disorder
- Frotteuristic disorder
- Paedophilic disorder
- Coercive sexual sadism disorder

Your answer is incorrect.

Frotteuristic disorder is characterised by a sustained, focused, and intense pattern of sexual arousal— as manifested by persistent sexual thoughts, fantasies, urges, or behaviours— that involves touching or rubbing against a non-consenting person in crowded public places. In addition, for it to be diagnosed, the individual must have acted on these thoughts, fantasies, or urges or be markedly distressed by them. It specifically excludes consensual touching or rubbing that occur with the consent of the person or persons involved.

The correct answer is: Frotteuristic disorder

Question 40

Not answered

Marked out of 1.00

When asked how many legs a cat has, a 42-year-old male prisoner answers 'five legs'. Which of the following conditions is likely?

Select one:

- Dementia
- Autistic Spectrum Disorder
- Schizophrenia
- Intellectual Disability
- Ganser syndrome

Your answer is incorrect.

Vorbeireden (also known as 'vorbeigehen' or 'talking past the point') is seen in Ganser syndrome. It refers to the production of 'approximate answers' in which the content of the patient's replies to questions shows that they understand what has been asked but have responded by talking about an associated topic.

Ganser syndrome is characterised by approximate answers, disorientation, clouding of consciousness, hallucinations/pseudo-hallucinations, motor disturbance, anxiety or apathy, normal ADLs, and sudden resolution with amnesia for the period of illness. In ICD-11, it is included under 'other specified dissociative disorders'.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30, Dieguez S. Ganser syndrome. Neurologic-Psychiatric Syndromes in Focus-Part II. 2018;42:1-22, Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019.

The correct answer is: Ganser syndrome

Question 41

Not answered

Marked out of 1.00

A middle aged man presents with 2 year history of having abnormal limb movements and memory problems. He has no past history of psychiatric problems and physical investigations are unremarkable. His father died at the age of 50 and he had similar 'limb shakes'. The likely diagnosis is

Select one:

- Wilson's disease
- Catatonic disorder
- Crutzfeld Jakob disease
- Huntington's disease
- Lewy body dementia

Your answer is incorrect.

Huntington's disease onset is defined by the beginning of motor symptoms, and most often the initial complaint that leads patients to seek medical attention is 'clumsiness', 'tremor', 'balance trouble', or 'jerkiness'. The primary involuntary movement abnormality, and often the earliest symptom, is chorea or choreoathetosis, continuous and irregular writhing and jerking movements. The limbs and trunk are most prominently affected.

Diagnosis: straightforward if patient with a known family history of HD presents with the typical neurological movement disorder (gene testing not required if there is laboratory confirmation of the Huntington's disease gene in the family); in a patient without a family history, gene testing is essential but does not give a clinical diagnosis and must of combined with clinical history and physical exam.

It results from expansion of a CAG trinucleotide repeat in the HTT (huntingtin) gene on the short arm of chromosome 4; alleles with 40 or more repeats are fully penetrant.

(Ref: Kumar A et al. Therapeutic Advances for Huntington's Disease. Brain Sciences. 2020 Jan;10(1):43, Bachoud-Lévi AC et al. International Guidelines for the treatment of Huntington's Disease. Frontiers in neurology. 2019;10:710).

The correct answer is: Huntington's disease

Question 42

Not answered

Marked out of 1.00

Which of the following is a characteristic feature of schizoid personality disorder?

Select one:

- Labile affect
- Insensitivity to prevailing social norms
- Highly sensitive and intolerability to setbacks
- Poor frustration tolerance
- Rigidity and stubbornness

Your answer is incorrect.

Taking pleasure in few, if any, activities, indifference to social norms and conventions, preoccupation with fantasy and introspection are features of schizoid personality.

Clinical features of patients with schizoid personality disorder include:

- Doesn't enjoy relationships
- Prefers doing things alone
- Not interested in sexual relations
- Doesn't enjoy activities
- Few if any friends or acquaintances
- Doesn't care what others think of them
- Emotionally constricted

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022.

The correct answer is: Insensitivity to prevailing social norms

Question 43

Not answered

Marked out of 1.00

According to DSM-5 criteria, which of the following is not included in the diagnostic criteria for a manic episode?

Select one:

- Increased energy levels
- Flight of ideas
- Much more talkative than usual
- Increased need for sleep
- Inflated self esteem/grandiosity

Your answer is incorrect.

As defined in DSM-5, a manic episode is a distinct period of abnormally and persistently elevated, expansive, or irritable mood and persistently increased activity or energy. To meet criteria, DSM-5 requires that three (or more) of the following symptoms (four if the mood is only irritable) are present to a significant degree and represent a noticeable change from usual behaviour:

- Inflated self-esteem or grandiosity
- Decreased need for sleep (e.g. feels rested after only 3 hours of sleep)
- More talkative than usual or pressure to keep talking
- Flight of ideas or subjective experience that thoughts are racing
- Distractibility
- Increase in goal-directed activity or psychomotor agitation
- Excessive involvement in activities that have a high potential for painful consequences

Ref: American Psychiatric Association. Diagnostic and statistical manual of mental disorders (DSM-5®). American Psychiatric Pub; 2013 May 22.

The correct answer is: Increased need for sleep

Question 44

Not answered

Marked out of 1.00

Which of the following diagnostic groupings includes selective mutism in ICD-11?

Select one:

- Neurodevelopmental disorders
- Obsessive-compulsive and related disorders
- Anxiety and fear-related disorders
- Dissociative disorders
- Disorders specifically associated with stress

Your answer is incorrect.

In ICD-11, Anxiety and fear-related disorders includes selective mutism and separation anxiety. Diagnoses are categorised by the focus of the apprehension for the individual, with this being diffuse in generalised anxiety disorder.

The correct answer is: Anxiety and fear-related disorders

Question 45

Not answered

Marked out of 1.00

A 76-year-old man presents to the memory clinic with a MMSE score of 23/30. What will you do in terms of management?

Select one:

- Wait and watch for 6 months
- Repeat memory test in 3 months time
- Prescribe antidepressant drugs
- Discharge back to GP
- Arrange for dementia screen work up

Your answer is incorrect.

MMSE score of 23/30 suggests a possibility of dementia. Consider dementia screen work up, which includes blood tests, urine tests, ECG and CT scan.

The correct answer is: Arrange for dementia screen work up

Question 46

Not answered

Marked out of 1.00

Physical findings that point to a metabolic cause of mental status change include

Select one:

- Hyporeflexia
- Positive Rhomberg's sign
- Positive Babinski sign
- Asterixis
- Dilated pupils

Your answer is incorrect.

It is helpful to look for asterixis, tremor and myoclonic jerks in patients who are confused and agitated. These are helpful signs that point to a metabolic or toxic cause of altered mental status. In the absence of myoclonus, asterixis, or other definitive localizing neurologic signs on an exam, the examiner should consider various conditions that could explain bihemispheric involvement.

Asterixis can be elicited by having the patient extend both arms with the wrists dorsiflexed and palms facing forward and eyes closed. Brief jerky downward movements of the wrist are considered a positive sign. Asterixis is commonly seen with metabolic encephalopathies.

Ref: Neurological History and Physical Examination - Medscape, http://emedicine.medscape.com/article/1147993-overview (accessed May 5, 2021), http://www.aan.com/go/education/curricula/internal/chapter4

The correct answer is: Asterixis

Question 47

Not answered

Marked out of 1.00

On examination, there is a repeated fluctuant movement of agonist and antagonist muscles when the patient tries to reach something. Where is the lesion?

Select one:

- Frontal lobe
- Basal ganglia
- Parietal lobe
- Pons
- Cerebellum

Your answer is incorrect.

The description refers to intentional tremor, which is a feature of cerebellar damage. Intentional tremors are the result of the lack of the cerebellum's ability to coordinate the timing of alternately contracting and inhibiting antagonistic muscles, and are often associated with dysdiadochokinesia.

The correct answer is: Cerebellum

Question 48

Not answered

Marked out of 1.00

Which one of the following eye signs is pathognomonic of multiple sclerosis?

Select one:

- Argyll Robertson pupil
- Kayser Fleischer ring
- Bilateral internuclear ophthalmoplegia
- Holmes Adie Pupil
- Horner's syndrome

Your answer is incorrect.

In young patients with bilateral internuclear ophthalmoplegia, multiple sclerosis is often the underlying cause. In older patients with one-sided ophthalmoplegia, stroke is a more common cause. Internuclear ophthalmoplegia (INO) strems from a dysfunction of the medial longitudinal fasciculus (MLF), a tract that contains axons projecting from the VI nucleus to the medial rectus subnuclei of the contralateral III nuclear complex. The cardinal findings in INO are impaired adduction and abducting nystagmus during conjugate version movements. The resulting ocular divergence leads to horizontal diplopia. (Ref: Barton JS (2008). Retrieved from http://www.neuroophthalmology.ca/textbook/disorders-of-eye-movements/)

The correct answer is: Bilateral internuclear ophthalmoplegia

Question 49

Not answered

Marked out of 1.00

A 64 year old lady has a history suggestive of frontal lobe dementia. Which of the following bedside cognitive tests is appropriate to assess frontal lobe function?

Select one:

- Rey Osterreith test
- Wisconsin card sorting test
- Visual field test
- Mini mental status examination
- Category fluency test

Your answer is incorrect.

Wisconsin card sorting test is not a bed side cognitive test. It is a more detailed neuropsychological assessment.

The correct answer is: Category fluency test

Question 50

Not answered

Marked out of 1.00

Intention tremor is characteristically seen in which of the following conditions?

Select one:

- Essential tremor
- Cerebellar disease
- Anxiety
- Lithium toxicity
- Frontal lobe disorders

Your answer is incorrect.

Intention tremor is an oscillating tremor that accelerates in pace on approaching the target. It is one of the neurological signs seen in disorders involving the cerebellum.

An intention (rubral, cerebellar, or course tremor) is defined as a rhythmic, oscillatory, and high amplitude tremor during a directed and purposeful motor movement, worsening before reaching the target. It is due to cerebellar dysfunction. In intention tremor, the central nervous system cerebellar feedback and error control centres are impaired. The relay systems between the cerebellum and efferent muscles can be compromised, resulting in the action tremor.

Ref: Rocha Cabrero F, De Jesus O. Intention Tremor. [Updated 2021 Jul 12]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2021 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK560642/

The correct answer is: Cerebellar disease

Question 51

Not answered

Marked out of 1.00

Which of the following is the single most useful tool for diagnosing dementia?

Select one:

- Clinical interview of patient and carer
- MMSE
- MRI scan
- Neuropsychological testing
- CT scan

Your answer is incorrect.

Nothing can replace a careful clinical interview for diagnostic evaluation of most disorders

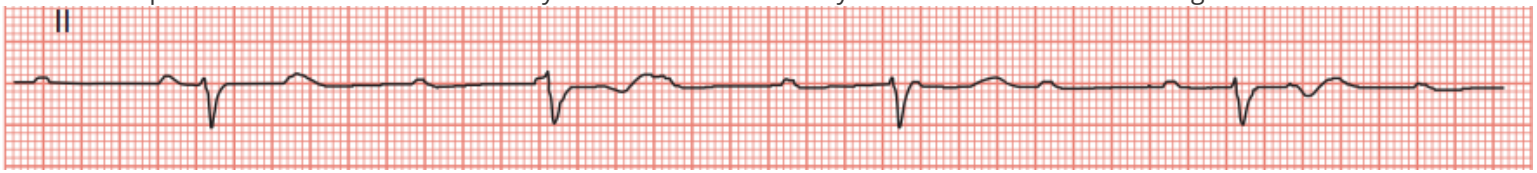
The correct answer is: Clinical interview of patient and carer

Question 52

Not answered

Marked out of 1.00

Read the strip of ECG shown below and identify the relevant abnormality. P waves are marked in the figure.



Select one:

- Mobitz type II
- Type I second degree heart block
- Mobitz type I
- Complete heart block
- Normal ECG

Your answer is incorrect.

There is a complete interruption of conduction between atria and ventricles so that the two appear to be working independently. The atrial P waves bear no relationship to the ventricular QRS complex, which usually arise as the result of a ventricular escape rhythm. (Ref: Houghton & Gray, Making sense of the ECG. Pg 120-122).

The correct answer is: Complete heart block

Question 53

Not answered

Marked out of 1.00

Amnesia refers to marked impairment in

Select one:

- Procedural memory
- Episodic memory
- All of the above
- Working memory
- Semantic memory

Your answer is incorrect.

Amnesia refers to a marked impairment in episodic memory, although other types of memory such as working memory, semantic memory and procedural memory may remain relatively intact. It is useful to remember two types of amnesia here; 1. Anterograde Amnesia: In hippocampal damage, a lack memory for the immediate future is seen. The subject cannot learn anything new as no information can be moved from STM to LTM. 2. Retrograde Amnesia: Following head injury, a lack memory for immediately preceding events is seen. The subject never consolidates the info (the information moves from STM to LTM, but cannot be retrieved when needed; thus the stored memory is 'lost' or inaccessible).

The correct answer is: Episodic memory

Question 54

Not answered

Marked out of 1.00

A 5-year-old girl loved to help her mother in kitchen. She was scared of going near the cooker after she burnt her hand. This is explained by

Select one:

- Reciprocal inhibition
- Extinction
- Negative reinforcement
- Positive reinforcement
- Operant conditioning

Your answer is incorrect.

When the conditioned stimulus no longer produces conditioned response, this is known as extinction. In this case, the extinction is associated with a 'punishing' experience of being burnt.

The correct answer is: Extinction

Question 55

Not answered

Marked out of 1.00

Which type of reinforcement is the least resistant to extinction?

Select one:

- Fixed interval
- Variable ratio.
- Continuous
- Fixed ratio
- Variable interval

Your answer is incorrect.

Reinforcement schedules may be continuous, fixed interval, fixed ratio, variable interval or variable ratio. Variable ratio reinforcement is the most resistant e.g., gambling. Continuous reinforcement is the one that can be easily made extinct. Intermittent reinforcement takes the longest to establish.

The correct answer is: Continuous

Question 56

Not answered

Marked out of 1.00

A psychologist takes a 31 year old patient with significant fear of heights to the top of a 21 storey-building. Which of the following behavioural technique is he employing?

Select one:

- Relaxation training
- Desensitisation
- Flooding
- Implosion
- In vitro flooding

Your answer is incorrect.

Flooding: It is an operant conditioning technique where actual exposure to feared stimulus takes place for a substantial amount of time, so the accompanying anxiety response fades away while the stimulus is continuously present. This will lead to an extinction of the fear. When a similar technique is attempted with imagined not actual exposure, then it is called as implosion.

The correct answer is: Flooding

Question 57

Not answered

Marked out of 1.00

In patients with moderate head injury, post traumatic amnesia is usually expected to last

Select one:

- From 1 week to 1 month
- From 1 day to 1 week
- Less than 1 hour
- From 1 to 24 hours
- Less than 1 minute

Your answer is incorrect.

Post traumatic amnesia is the time between the injury and recovery of normal continuous memory, seen in head injury patients. The longer the PTA, the more severe the brain damage and poorer the prognosis for the recovery. PTA grading system is as follows; PTA less than 1 hour- mild head injury PTA from 1 to 24 hours-moderate head injury PTA longer than 24 hours-severe injury. Retrograde amnesia is also possible after a head injury. In most cases, the amnesic gap is short and lasts for less than one minute. It is not a good indicator of prognosis.

The correct answer is: From 1 to 24 hours

Question 58

Not answered

Marked out of 1.00

Which of the following statements regarding operant conditioning is correct?

Select one:

- It cannot be used in patients with learning disability as the concept of reward is non-existent in most patients
- Punishment is the most effective form of operant learning
- It is a form of associative learning
- It was first described by Albert Bandura
- Negative reinforcement refers to introducing an aversive stimulus to reduce the frequency of a behaviour

Your answer is incorrect.

Operant conditioning and classical conditioning are two types of associative learning. Habituation and sensitisation are described as non-associative learning. Punishment is not an effective conditioning technique; it helps to learn how not to behave but not how to behave. On most occasions, it only serves to suppress a response - not abolishing it completely.

The correct answer is: It is a form of associative learning

Question 59

Not answered

Marked out of 1.00

Which statement among the following about short-term memory is false?

Select one:

- The retrieval process in short term memory is error free.
- Information is held in the STM for 15 to 30 seconds.
- It can be extended by rehearsal
- Chunking of material can effectively increase the capacity of short term memory
- It is mainly stored in the visual modality

Your answer is incorrect.

The information in the short-term memory is mainly stored in the auditory modality. In STM, the encoding is acoustic, and the retrieval is error-free.

The correct answer is: It is mainly stored in the visual modality

Question 60

Not answered

Marked out of 1.00

Which of the following is a behavioural component of Allport's concept of prejudice?

Select one:

- Circumlocution
- Stereotype
- Discrimination
- Hostility
- Moral attack

Your answer is incorrect.

Prejudice is essentially an attitude. Hence, it has 1. Cognitive component - stereotypes; 2. Affective component - hostility; 3. Behavioural component - which according to Allport can be of several types in terms of increasing severity starting from anti-locution, avoidance, discrimination, physical attack and in some cases, ending in extermination .

The correct answer is: Discrimination

Question 61

Not answered

Marked out of 1.00

A police officer arrives at a place to investigate a reported crime. He notices a suspicious looking man who looks at the officer and runs way without talking to the officer. The police team arrests him out of suspicion and is not ready to listen to any of his explanations. But he turns out to be not guilty as the investigation proceeds. Which of the following is a likely explanation?

Select one:

- Primacy effect
- Hawthorne effect
- Halo effect
- Barnum effect
- Pygmalion effect

Your answer is incorrect.

Halo effect is the tendency to perceive other persons as wholly good or bad based on few observed traits (e.g. physical attractiveness). Thus a person's positive or negative traits 'spill over' from one area to influence the total perception of their personality. Investigators evaluating suspects are susceptible for halo effect (to be accurate - reversed halo effect or devil effect or association fallacy). For example, a policeman may conclude someone is guilty by association with attributes he has previously seen in other criminals. Mere similarity of a person to a suspect often causes the police to wrongly associate the innocent with a guilty act.

The correct answer is: Halo effect

Question 62

Not answered

Marked out of 1.00

Which of the following instruments could be used to measure the discriminant attitudes and level of stigma in a community?

Select one:

- Changing minds scale
- Helsinki Scale of Stigma Burden
- Community attitudes to mental illness scale
- Openness and discrimination scale
- Stigma and Social prejudice scale

Your answer is incorrect.

Community Attitudes to Mental Illness (CAMI) measures attitudes across a number of key parameters. It was developed specifically for use in the community (in the context of the emptying of the asylums) and refined to quantify attitudes across four scales: authoritarianism, benevolence, social destructiveness and community mental health ideology. It was developed by Taylor and Dear in 1981 (Psychiatric stigma. The British Journal of Psychiatry, http://bjp.rcpsych.org/content/178/3/281).

The correct answer is: Community attitudes to mental illness scale

Question 63

Not answered

Marked out of 1.00

According to Holmes and Rahe's social readjustment rating scale (SRRS) which one was rated as the most serious life event?

Select one:

- Death of a spouse
- Loss of a job
- Violation of law
- Moving home
- Marriage

Your answer is incorrect.

Holmes and Rahe created a list of 43 life events of carrying severity collected from the analysis of data from 5000 patient records. Death of a spouse was arbitrarily rated as the most serious life event with minor violations of law rated as the least serious.

The correct answer is: Death of a spouse

Question 64

Not answered

Marked out of 1.00

Koro can be classified as a/an

Select one:

- Somatisation phenomenon
- Autoscopic phenomenon
- Desomatisation phenomenon
- Heautoscopic phenomenon
- Derealisation phenomenon

Your answer is incorrect.

Koro is a culture-bound syndrome. The fear of shrinking genitalia seen in Koro can be phenomenologically classified as a desomatisation phenomenon.

The correct answer is: Desomatisation phenomenon

Question 65

Not answered

Marked out of 1.00

A Sri Lankan man presents with depression and fatigue, and attributes it to excessive masturbation and losing semen in urine. What is the name of his condition?

Select one:

- Koro
- Susto
- Dhat
- Latah
- Piblokto

Your answer is incorrect.

This is called as Dhat syndrome. It is seen in India and some parts of South East Asia. Dhat refers to the severe anxiety and hypochondriacal concerns associated with the seminal discharge accompanied by feeling weak and exhausted. Patients may show guilt for excessive masturbation.

The correct answer is: Dhat

Question 66

Not answered

Marked out of 1.00

At what age can a child build a tower of 6 cubes?

Select one:

- 2 years
- 6 -10 months
- 5 years
- 12-18 months
- 4 years

Your answer is incorrect.

At age 2, a child can run and build a tower of 6 cubes. At age 3, a child can make a tower of 9-10 cubes.

The correct answer is: 2 years

Question 67

Not answered

Marked out of 1.00

Mastering intimate relationships despite the demands of work is a successful developmental task for ages between

Select one:

- 60 to 80
- 40 to 60
- 20 to 40
- 10 to 15
- 80 to 90

Your answer is incorrect.

The Intimacy vs. Isolation conflict is emphasized around the ages of 30. At the start of this stage, identity vs. role confusion is coming to an end, though it still lingers at the foundation of the stage (Erikson, 1950).

The correct answer is: 20 to 40

Question 68

Not answered

Marked out of 1.00

Which of the following is correct with regard to day-care for children?

Select one:

- 4 months day care before age 1 for more than 20hours/week may be detrimental
- Day care reduces attachment behaviour
- Day care induces neurotic clinging
- Regular day care affects cognitive development
- Day care increases maternal bonding

Your answer is incorrect.

A series of studies by Belsky and Rovine (1988) claim that insecure attachments were more likely to develop if the child had been receiving care of at least 20 hours per week for 4 months or more before its first birthday.

The correct answer is: 4 months day care before age 1 for more than 20hours/week may be detrimental

Question 69

Not answered

Marked out of 1.00

Which of the following antidepressants is associated with a significant risk of agranulocytosis compared to the others listed?

Select one:

- Mirtazapine
- Bupropion
- Escitalopram
- Fluoxetine
- Venlafaxine

Your answer is incorrect.

All drugs ending with a suffix -zapine can cause agranulocytosis (mirtazapine, olanzapine, clozapine, carbamazepine, etc.). In addition, mianserin can also cause this effect

The correct answer is: Mirtazapine

Question 70

Not answered

Marked out of 1.00

Which of the following antidepressants has a recognised side effect of elevated cholesterol?

Select one:

- fluoxetine
- imipramine
- venlafaxine
- paroxetine
- mirtazapine

Your answer is incorrect.

Mirtazapine - similar to olanzapine - is associated with blood dyscrasias and hyperlipidemia.

The correct answer is: mirtazapine

Question 71

Not answered

Marked out of 1.00

Which of the following is NOT a dystonia?

Select one:

- Writer's cramp
- Oculogyric crisis
- Motor tic
- Torticollis
- Blepharospasm

Your answer is incorrect.

Tics are not classified as dystonia

The correct answer is: Motor tic

Question 72

Not answered

Marked out of 1.00

Veronica has been going through a period of change in her life, and she's feeling particularly stressed. She is finding it especially difficult to get to sleep. When you take a careful history, you discover she's been drinking three cups of coffee before going to bed. Which is the main neuro-receptor implicated in Veronica's struggle to get to sleep after drinking a lot of coffee?

Select one:

- Adenosine receptor
- Serotonin receptor
- Dopamine receptor
- GABA receptor
- Opioid receptor

Your answer is incorrect.

Coffee's stimulant action is largely due to its action on adenosine receptors. When adenosine binds to its receptors, neural activity slows down and this is associated with feeling sleepy. Caffeine is an adenosine receptor antagonist and has the effect of speeding up neural activity and making us feel more alert.

The correct answer is: Adenosine receptor

Question 73

Not answered

Marked out of 1.00

Shaun is an 11-year-old child who has recently been diagnosed with epilepsy. He has been prescribed a medication to reduce the incidence of his seizures, but his parents are reporting that he has had some angry outbursts, behaving in a destructive manner, smashing some of his toys and frightening his siblings. Which of the following anti-epileptics is most likely to be associated with behavioural disturbances?

Select one:

- Vigabatrin
- Gabapentin
- Valproate
- Pregabalin
- Lamotrigine

Your answer is incorrect.

Of the medications listed, Vigabatrin is most associated with behavioural disturbances, particularly in children; a French study found behavioural disturbances in children taking Vigabatrin to be as high as 26% (Ref: Epilepsia Vol 47 Suppl 2, 2006).

The correct answer is: Vigabatrin

Question 74

Not answered

Marked out of 1.00

Carol has been taking Paroxetine 40mg every day for the past two months in the hope that it may alleviate her feelings of depression. She thinks her mood has improved somewhat, and now she wishes to stop the antidepressant. She doesn't consult her GP, but she stops her medication abruptly. Over the next few days she feels unwell: tired, with aching muscles and joints, headaches and nauseated. The feeling pass after about a week. She reads up on why this might have happened and she thinks it might be to do with SSRI discontinuation syndrome. What is a proposed mechanism for SSRI Discontinuation Syndrome?

Select one:

- Temporary deficiency in the brain of one or more essential neurotransmitters associated with mood
- Temporary excess in the brain of one or more essential neurotransmitters associated with mood
- Long-standing changes to brain structure which reduce the brain's capacity to produce neurotransmitters associated with mood
- Release of inflammatory proteins producing flu-like symptoms
- There is no physiological change, but it is all explained in psychological terms

Your answer is incorrect.

Even though SSRI Discontinuation Syndrome is not completely understood, this answer is the most likely according to current research in the area. Neurotransmitters implicated include serotonin, dopamine and acetylcholine (cholinergic rebound). (Ref: Am Fam Physician 2006 1;74(3))

The correct answer is: Temporary deficiency in the brain of one or more essential neurotransmitters associated with mood

Question 75

Not answered

Marked out of 1.00

Paul has been taking Risperidone as recommended by her psychiatrist, following an admission to hospital during which she was experiencing auditory hallucinations. She has now noticed that her periods have stopped, and there is milk leaking from her breasts. Her blood investigations reveal a Prolactin level of 960 MiU/L. What dopamine receptor is most implicated in raised prolactin?

Select one:

- D2
- D1
- D3
- D5
- D9

Your answer is incorrect.

High affinity to D2, and slow dissociation from the receptor once bound, is associated with high prolactin levels. Dopamine itself has a central role in the regulation of prolactin secretion, a mechanism that is inhibited by several antipsychotics.

The correct answer is: D2

Question 76

Not answered

Marked out of 1.00

Antonio has had poorly controlled diabetes since he was 5 years old. Now in his early 20s, he takes Captopril in order to control his hypertension, which has resulted from renal dysfunction. He is admitted to a psychiatric ward in a state of disinhibition, and appears elated, grandiose and psychotic. You are considering starting him on a medication to stabilise his mood. Which medication should be avoided in particular if we take into consideration his physical health problems and current medication?

Select one:

- Lithium
- Soldium Valproate
- Lamotrigine
- Olanzapine
- Risperidone

Your answer is incorrect.

Lithium is renally excreted and must be used with extreme caution in patients with impaired renal function, and in conjunction with medications such as ACE inhibitors, which may affect renal function. In such cases there's the risk that Lithium is not excreted at an adequate rate, leading to toxicity.

The correct answer is: Lithium

Question 77

Not answered

Marked out of 1.00

Mrs Myrtle has recently been discharged from a two-month admission to hospital. She was diagnosed with schizophrenia ten years ago and her concordance with oral medication has been poor. On this last admission she was started on a depot medication. At a follow-up meeting she is accompanied by her husband who reports that since this last admission she sometimes kicks him in bed. She also doesn't seem to be able to sit still in a chair. What is the best fitting description?

Select one:

- Akathisia
- Aggressive disorder
- Tardive dyskinesia
- Parkinsonism
- REM sleep disorder

Your answer is incorrect.

Antipsychotics can precipitate akathisia, a movement disorder characterised by restlessness, fidgeting and a difficulty in keeping still.

The correct answer is: Akathisia

Question 78

Not answered

Marked out of 1.00

Mr Farah was admitted to Psychiatric Intensive Unit in an acutely psychotic state, believing the MI5 to be persecuting him, and resulting in him lashing out at strangers in public places, causing physical injury. He is restrained and given an injection of an antipsychotic medication. Two hours later, when he is re-examined, he's found to be pyrexic, hypertensive, confused, with muscle rigidity. What is the most likely cause of this presentation?

Select one:

- Neuroleptic Malignant Syndrome
- HIV Encephalitis
- Heamorrhagic stroke
- Serotonin Syndrome
- Anticholinergic syndrome

Your answer is incorrect.

NMS is a rare but life-threatening reaction to neuroleptic medications characterised by muscle rigidity, fever, autonomic dysregulation, altered mental state.

The correct answer is: Neuroleptic Malignant Syndrome

Question 79

Not answered

Marked out of 1.00

Mr Samuel Hicks takes a regular antipsychotic medication and has monthly blood monitoring for this. He presents acutely with fever and chest pain. What ECG changes would you see?

Select one:

- Saddling of the ST segment
- Prolongation of PR segment
- Tenting of T waves
- Evidence of deep Q waves
- Atrial bigeminy

Your answer is incorrect.

Mr Hicks will be taking Clozapine as he is having regular blood tests for an antipsychotic prescription. This can cause myocarditis as a rare, but serious, side effect. This is characterised by saddled elevation of the ST segment, diffuse T wave inversion and other non-specific ST changes.

The correct answer is: Saddling of the ST segment

Question 80

Not answered

Marked out of 1.00

Sally's father died of Huntington's disease when Sally was 13. Now aged 33, she has decided to get tested because she has noticed that her limbs are making unusual movements, beyond her control. She tests positive for Huntington's. Five years later her movement problems have become significantly worse and are affecting her functioning. Which medication is she most likely to be prescribed to help with her Huntington's related movement disorder?

Select one:

- Tetrabenazine
- Risperidone
- Citalopram
- Levodopa
- Diazepam

Your answer is incorrect.

Tetrabenazine is an approved drug for this condition. It acts as a dopaminergic antagonist, but also has effects on other monoamines, leading to their reversible depletion. It may also be used for other hyperkinetic disorders such as Tourette's syndrome and Tardive Dyskinesia.

The correct answer is: Tetrabenazine

Question 81

Not answered

Marked out of 1.00

Mr Brown has had alcohol dependence for the past three decades. His daughter brought him to the A&E in an agitated state. The A&E staff have arranged for a set of bloods to be taken, including liver function tests. You are informed that his daughter gave him 10mg of Diazepam (her own prescription) to help calm her father down before he arrived in A&E. As part of your onward management plan, you wish to take into consideration the protein binding and metabolic properties of Diazepam. Which one of the following statements is true:

Select one:

- Diazepam has protein binding of 98% with the main active metabolite being desmethyldiazepam
- Diazepam has protein binding of 98% with temazepam as the main active metabolite
- Diazepam has protein binding of 98% with no active metabolites
- Diazepam has protein binding of 65% with the main active metabolite being desmethyldiazepam
- Diazepam has protein binding of 65% and no active metabolites

Your answer is incorrect.

Diazepam has protein binding of 98% with the main active metabolite being desmethyldiazepam. Minor active metabolites are Oxazepam and Temazepam

The correct answer is: Diazepam has protein binding of 98% with the main active metabolite being desmethyldiazepam

Question 82

Not answered

Marked out of 1.00

The pseudocholinesterase deficiency is associated with increased sensitivity to which of the following chemicals?

Select one:

- Donepezil
- Sodium Valproate
- Fluoxetine
- Carbamazepine
- Propofol

Your answer is incorrect.

It is advised that Donepezil is avoided in people with confirmed pseudocholinesterase deficiency. There are case reports of adverse effects, particularly when certain muscle relaxants (Succylcholine and Mivacurium) are used.

The correct answer is: Donepezil

Question 83

Not answered

Marked out of 1.00

Caffeine is a commonly used and socially acceptable psychoactive substance. Which of the following statements is incorrect regarding caffeine?

Select one:

- Saliva concentration is half of plasma level
- It is metabolised by CYP1A2?
- It is more damaging to foetus in rapid metabolisers than slow.
- It can cross the placenta?
- It is metabolised similarly in all ethnic groups?

Your answer is incorrect.

Caffeine is present in serum and saliva is similar concentrations. Caffeine is primarily metabolised by CYP1A2 and also, in smaller amounts, by CYP2E1, CYP3A4, CYP2C8, CYP2C9. Konje and Cade (BMJ - 2008) showed the association between caffeine and foetal growth restriction was stronger in women with a faster compared to a slower caffeine clearance. It crosses the placenta freely. There are ethnic differences between the speed of caffeine metabolism.

The correct answer is: Saliva concentration is half of plasma level

Question 84

Not answered

Marked out of 1.00

The blood brain barrier has an important role in only permitting selected molecules through it and into the cerebral and cerebellar tissue. Which of the following does it allow through?

Select one:

- L-DOPA
- Lipophilic big un-ionised molecules
- Hydrophilic molecules
- Histamine
- Noradrenaline

Your answer is incorrect.

L-DOPA can pass through the blood brain barrier via a specialized transmembrane transporter called L-type amino acid transporter. Molecules can pass through the blood brain barrier if they are small, lipophilic (un-ionised), a gas or if they have a specific transporter.

The correct answer is: L-DOPA

Question 85

Not answered

Marked out of 1.00

Serotonin is thought to be an important neurotransmitter in the aetiology and treatment of mental disorder. Which of the following enzymes are not involved in its metabolism?

Select one:

- COMT
- MAO
- Aldehyde dehydrogenase
- DOPA decarboxylase
- Tryptophan hydroxylase

Your answer is incorrect.

COMT (Catechol-O-methyltransferase) is one of several enzymes that degrade catecholamines (such as dopamine, adrenaline, and noradrenaline), catecholestrogens, and various drugs and substances having a catechol structure. Serotonin does not have a catechol structure (benzene with two hydroxyl side groups at carbons 1 and 2) and a side-chain amine.

The correct answer is: COMT

Question 86

Not answered

Marked out of 1.00

Which of the following is acceptable to be taken along with MAO inhibitors?

Select one:

- Mature cheese
- Broad bean pods
- Red wine
- Pickled Herring
- Grapefruit juice

Your answer is incorrect.

Except grapefruit juice, the others on the list are rich in tyramine and so can produce cheese reaction.

The correct answer is: Grapefruit juice

Question 87

Not answered

Marked out of 1.00

Which routes of administration leads to nicotine reaching the brain fast?

Select one:

- Sublingual
- Inhalational
- Intravenous
- Cutaneous
- Oral

Your answer is incorrect.

Drugs administered via the inhalational route reaches the brain quickly.

The correct answer is: Inhalational

Question 88

Not answered

Marked out of 1.00

Which one of the following statements about reboxetine is correct?

Select one:

- It is highly cardiotoxic.
- Reboxetine is a hepatic enzyme inhibitor
- Reboxetine is a hepatic enzyme inducer
- It does not affect hepatic cytochrome P450 enzymes.
- It is completely excreted via kidneys

Your answer is incorrect.

Reboxetine is rapidly absorbed after oral doses. It is mainly metabolised in the liver and has an elimination half-life of about 13 hours. In clinical trials, reboxetine has been shown to be as effective as the SSRIs in the treatment of depression but, unlike the SSRIs, reboxetine does not inhibit any of the cytochrome P450 enzymes in the liver.

The correct answer is: It does not affect hepatic cytochrome P450 enzymes.

Question 89

Not answered

Marked out of 1.00

An orally administered medication requires five half-life periods to reach a steady state concentration. Which of the following statements is most likely to be correct with respect to this drug?

Select one:

- Doses up to five times the normal are non-toxic
- It follows first order kinetics
- A loading dose must be provided for adequate therapeutic activity
- Its rate of metabolism is independent of its concentration
- This medication must be administered only once a day

Your answer is incorrect.

Under conditions of first-order kinetics, given that a drug is rapidly and evenly distributed throughout the body and no loading dose is given, at least 5 half-lives are required to achieve a steady state concentration. Drugs following zero order kinetics show a concentration-independent elimination.

The correct answer is: It follows first order kinetics

Question 90

Not answered

Marked out of 1.00

Which of the following is true with regard to sodium valproate?

Select one:

- Intravenous preparation is available for acute mania
- It is excreted unchanged via kidney
- It has no significant interactions with other mood stabilisers
- Should not be combined with antidepressants
- It increases carbamazepine levels

Your answer is incorrect.

Valproate appears to inhibit the glucuronidation of carbamazepine, leading to an increase in its peak levels. This may be associated with the clinically significant dose related effects of carbamazepine such as ataxia. Interactions are also seen with other psychotropics (especially lamotrigine). A study involving the administration of 100 to 300 mg/ day of chlorpromazine to schizophrenic patients already receiving valproate (200 mg BID) revealed a 15% increase in trough plasma levels of valproate.

The correct answer is: It increases carbamazepine levels

Question 91

Not answered

Marked out of 1.00

Regarding pharmacokinetics and properties of flumazenil, which of the following is true?

Select one:

- It is an inverse agonist at GABA receptor
- It has an elimination half-life of 4 hours
- It reduces panic attacks instantaneously
- It is an epileptogenic drug
- It is a partial agonist at GABA receptor

Your answer is incorrect.

Flumazenil is a competitive antagonist of benzodiazepines at benzodiazepine sites of GABA-A receptors. Hence it is an epileptogenic drug. It is an imidazobenzodiazepine derivative with weak partial agonist action in some animal models but has no agonist activity in humans. By blocking GABA A neurotransmission, it can induce panic attacks. This has some diagnostic value too. It has a very short elimination t O and so must be given as a continuous infusion. It is 40 to 50 percent protein bound with significant tissue uptake. It is primarily metabolized by the liver with t1/2 of 40 - 80 minutes.

The correct answer is: It is an epileptogenic drug

Question 92

Not answered

Marked out of 1.00

Which of the following is hydroxylated as one of the steps in dopamine synthesis?

Select one:

- Tryptophan hydroxylase
- L-tyrosine
- L-dopa
- L-tryptophan
- Phenylalanine hydroxylase

Your answer is incorrect.

Tyrosine is converted to dopamine by the enzymes tyrosine hydroxylase (TH) and L-amino acid decarboxylase (AADC) also called dihydroxyphenylalanine (DOPA) decarboxylase (DDC).

The correct answer is: L-tyrosine

Question 93

Not answered

Marked out of 1.00

A 63-year-old man has a long-standing history of mood disorder. He is also taking Ramipril for hypertension. Which of the following drugs is most likely to cause a possible interaction?

Select one:

- Venlafaxine
- Amitriptyline
- Fluoxetine
- Lithium
- Bupropion

Your answer is incorrect.

Ramipril is an Angiotensin converting enzyme inhibitors (ACEI). ACEI can reduce thirst, which can lead to mild dehydration, and increase renal sodium loss, leading to increased sodium reabsorption by the kidney causing an increase in lithium plasma levels and, therefore, precipitate lithium toxicity. ACE inhibitors can also precipitate renal failure. Therefore, if ACEI are co-prescribed with lithium, more frequent monitoring of e-GFR and plasma lithium is required. (Ref: The Maudsley prescribing guidelines- 11th edition-Pg. 164)

The correct answer is: Lithium

Question 94

Not answered

Marked out of 1.00

Which of the following is correct with regard to the pharmacology of lithium?

Select one:

- Lithium is unsafe in the elderly
- Lithium is contraindicated in patients with sick sinus syndrome
- Obesity is associated with lower rates of lithium clearance
- It is not excreted in breast milk
- Lithium cardiotoxicity is more prevalent in people on a high-salt diet

Your answer is incorrect.

Lithium is safe and effective in the elderly; it is excreted in the breast milk and must be carefully considered in lactating mothers.BMI does not affect lithium clearance as long as renal function is normal; Lithium can cause a shift in intracellular potassium and is contraindicated in sick sinus syndrome. The cardiotoxicity is more common in those on low salt diet and those with renal insufficiency.

The correct answer is: Lithium is contraindicated in patients with sick sinus syndrome

Question 95

Not answered

Marked out of 1.00

What part of the Weschler's Adult Intelligence Scale is most likely to be resistant to brain damage?

Select one:

- Digit Symbol
- Digit Span
- Picture completion
- Object assembly
- Matrix assembly

Your answer is incorrect.

Tests on WAIS are either 'hold' or 'no hold'. The so called 'hold tests' in WAIS are thought to reflect the use of old knowledge and are relatively resistant to the effects of brain damage and ageing. These include vocabulary, information, object assembly, and picture completion. 'No hold' tests require speed of response, working memory, or the creation of new relations between unrelated items. These are more likely to show an early decline with ageing and cognitive impairment. These include digit symbol, digit span, similarities, and block design. This explains why WAIS is not the best test to measure premorbid intelligence. A deterioration index is calculated using the composite scores of HOLD and NON-HOLD subtests.

The correct answer is: Picture completion

Question 96

Not answered

Marked out of 1.00

A 43-year-old man with pure word deafness is likely to show which of the following?

Select one:

- Reading is unimpaired
- Comprehension of speech is intact
- Writing is impaired
- Meaning of words is intact when asked
- All of the above

Your answer is incorrect.

Pure word deafness: Patient can speak read & write fluently, but comprehension is impaired only for spoken language. Bilateral (or left sided with disrupted connections to non-dominant circuit) damage to the superior temporal pole is suspected. In Wernicke's aphasia, the comprehension is impaired for both spoken and written language.

The correct answer is: Reading is unimpaired

Question 97

Not answered

Marked out of 1.00

Which one of the following tests are not correctly linked with the main function tested?

Select one:

- Benton test- Visual memory.
- Paired associates test-verbal memory
- Stroop test-executive functioning
- Halstead-Reitan test- Intelligence
- Rey-Osterreith test- To detect prefrontal lobe pathology

Your answer is incorrect.

Rey-Osterreith test measures the recall of learned material. It is a test of visual memory. Paired associate test tests for verbal memory using paired and unpaired words. Wisconsin card sorting test is used to detect prefrontal lobe pathology.

The correct answer is: Rey-Osterreith test- To detect prefrontal lobe pathology

Question 98

Not answered

Marked out of 1.00

There is a time gradient notable when patients develop retrograde amnesia, so that recent memories are more likely to be lost than the more remote memories. This is called

Select one:

- Interference theory
- Ribot's Law
- Law of effect
- Yerkes Dodson law
- Decay theory

Your answer is incorrect.

Ribot in 1881 wrote a book (Les Maladies de la Memoire (The Diseases of Memory)) in which he claimed on the basis of clinical observations, that brain injury affects premorbid memories in the reverse order of their formation such that newly formed memories (i.e., those formed just prior to brain injury) are impaired to a greater extent than older memories. Today, Ribot's law is referred to as temporally graded retrograde amnesia, and much evidence involving both human and animal subjects supports its validity (Ref: Wixted, J. T. (2004). On Common Ground: Jost's (1897) law of forgetting and Ribot's (1881) law of retrograde amnesia. Psychological review, 111(4), 864.)

The correct answer is: Ribot's Law

Question 99

Not answered

Marked out of 1.00

The most obviously affected neuropsychological measure in those with post concussion syndrome is

Select one:

- Speed of processing
- Procedural memory
- Block designing
- Verbal fluency
- Picture matching

Your answer is incorrect.

Processing speed is the most obviously affected neuropsychological measure in those with post concussion syndrome.

The correct answer is: Speed of processing

Question 100

Not answered

Marked out of 1.00

A patient has with UMN lesion of cranial nerves 9, 10, 11 and 12. Which of the following dysfunction is likely to be seen?

Select one:

- Facial droop
- Tongue fasciculations
- Swallowing problems
- Chewing problems
- Loss of touch sensation on left side of face

Your answer is incorrect.

Swallowing - IX and X; Chewing and facial sensation: - V. Facial muscle tone- VII. Fasciculations are seen in LMN lesion

The correct answer is: Swallowing problems

Question 101

Not answered

Marked out of 1.00

You want to measure the impairment of everyday memory functioning to facilitate rehabilitation of individuals with acquired brain damage. The most suitable test is

Select one:

- Wechsler memory scale
- Halstead-Reitan Neuropsychological Battery
- The Rivermead behavioral memory test
- Ravens progressive matrices
- Thematic apperception test

Your answer is incorrect.

The Rivermead Behavioral Memory Test detects impairment of everyday memory functioning and also monitors changes following the treatment for memory difficulties. The aim of the test is to provide task analogues of everyday memory problems that appear challenging for patients with memory impairment. It can be used in adults with memory problems; individuals with traumatic or acquired brain injury, dementia, or MS.

The correct answer is: The Rivermead behavioral memory test

Question 102

Not answered

Marked out of 1.00

A 45-year-old gentleman has history of alcohol dependence. He has no memeory problems or neurological signs but appears disorganised in day to day tasks. If a neuropsychologist tests him he is most likely to show impaired performance in which of the following tests?

Select one:

- Orientation
- Clock drawing test
- Trail making tests
- Digit span
- Picture completion test

Your answer is incorrect.

Trial making tests is a test of frontal lobe function. Patients with alcohol dependence have decrements in executive function (frontal lobe), and it could be used.

The correct answer is: Trail making tests

Question 103

Not answered

Marked out of 1.00

A 76-year-old man was brought to Emergency Unit by his wife. He had breakfast that morning and appeared unable to remember that he and his wife have planned a day out. He repeatedly enquired where she was taking him. She reports that he appeared bewildered, but was alert and had no sensory and motor deficits throughout. He returned to normal after 10 hours. What is the most likely diagnosis?

Select one:

- Creutzfeldt-Jakob disease
- Transient global amnesia
- Alzheimer's disease
- Normal-pressure hydrocephalus
- Parkinson's disease

Your answer is incorrect.

Transient global amnesia (TGA) refers to an episode of complete and reversible anterograde and retrograde memory loss lasting up to 24 h. Patients have a persistent loss of memory for the time of the attack. During the episode, patients often appear bewildered and ask repeated questions.

The correct answer is: Transient global amnesia

Question 104

Not answered

Marked out of 1.00

Which of the following cannot be termed as a reliable endophenotype for schizophrenia?

Select one:

- Glial cell changes
- Working memory defects
- Prepulse inhibition
- Smooth pursuit defects
- Negative symptoms

Your answer is incorrect.

The suggested criteria that define an endophenotype include: the marker is associated with the illness in the population, heritable, state-independent, co-segregates with the illness within families and is present at a higher rate in unaffected family members of the proband compared with the general population. Clinical symptom clusters do not fit these criteria.

The correct answer is: Negative symptoms

Question 105

Not answered

Marked out of 1.00

Which one among the following is the least likely statement about Huntington's disease?

Select one:

- Huntington genes code for production of protein called huntingtin
- CAG repeat size and age are inversely correlated
- Late onset Huntington's disease is usually inherited from mother
- Paternally derived gene may be more unstable than maternally derived gene
- Normal Huntingtin gene has 35-45 CAG repeats

Your answer is incorrect.

Huntington's disease is a fully penetrant autosomal disorder. The affected gene, at 4p16, codes for huntingtin. In Huntington's disease, there are increased CAG repeats. Normal repeat lengths range from about 7 to 28 triplets. Individuals with 29-35 triplets will not develop HD but may pass an expanded allele to an offspring, while individuals with 40 or more triplets will develop HD. (Ref: New Oxford Textbook- 372)

The correct answer is: Normal Huntingtin gene has 35-45 CAG repeats

Question 106

Not answered

Marked out of 1.00

Where does transcription take place in human cells?

Select one:

- Rough endoplasmic reticulum
- Smooth endoplasmic reticulum
- Cytoplasm
- Golgi cmplex
- Nucleus

Your answer is incorrect.

Active transcription units are clustered in the nucleus, in discrete sites called transcription factories or euchromatin.

The correct answer is: Nucleus

Question 107

Not answered

Marked out of 1.00

Probandwise concordance rate is calculated as

Select one:

- Number of total pairs divided the affected population
- Number of twin pair divided by the total number of sib pairs
- Number of affected twins divided by total number of co-twins
- Number of twin pairs divided by the total relatives
- Number of twin pair divided by the total population

Your answer is incorrect.

A pairwise concordance rate is estimated as the number of twin pairs who both have the disorder divided by the total number of pairs. However, where there has been systematic ascertainment, one can report a probandwise concordance rate that is calculated as the number of affected twins divided by the total number of cotwins.

The correct answer is: Number of affected twins divided by total number of co-twins

Question 108

Not answered

Marked out of 1.00

Which of the following gene, when knocked out, produces mice with phenotype resembling narcolepsy?

Select one:

- Neuropeptide Y
- Vasopressin
- Orexin
- Substance P
- Hydrocortisone

Your answer is incorrect.

Hypocretin-1 and -2 (also called orexin-A and -B) are newly discovered neuropeptides processed from a common precursor, preprohypocretin. Hypocretin-containing cells are located exclusively in the lateral hypothalamus, with widespread projections to the entire neuraxis.The discovery that hypocretins are involved in narcolepsy, a disorder associated with excessive daytime sleepiness, cataplexy and unusually rapid transitions to rapid-eye-movement sleep, opens a new field of investigation in the area of sleep control physiology (Hypocretin/orexin, sleep and narcolepsy, http://www.ncbi.nlm.nih.gov/pubmed/11340621).

The correct answer is: Orexin

Question 109

Not answered

Marked out of 1.00

The technique used to assemble correct DNA fragments from a gene library into a continuous overlapping stretch, which will contain genes of interests is;

Select one:

- Physical mapping
- Polymerase chain reaction
- Association studies
- Mutagenesis
- Linkage analysis

Your answer is incorrect.

Physical maps of DNA give the physical, DNA-base-pair distances from one landmark to another. By using the overlapping DNA sequences at the ends of the genome pieces to keep track of where the pieces came from, physical mapping is accomplished.

The correct answer is: Physical mapping

Question 110

Not answered

Marked out of 1.00

What is the best test for diagnosing Huntington's disease?

Select one:

- CT scan
- Serum Protein Assay
- Polymerase Chain Reaction
- Karyotyping
- MRI scan

Your answer is incorrect.

Serum polymerase chain reaction (PCR) is the test of choice to examine the number of trinucleotide repeats (More than35 in adults and More than50 in children) in order to diagnose Huntington's disease (HD). The causative gene resides on the short arm of chromosome 4 at 4p16.3. A chromosomal karyotype can reveal only macroscopic defects in chromosomes such as deletions, translocations, or trisomies.

The correct answer is: Polymerase Chain Reaction

Question 111

Not answered

Marked out of 1.00

Polyploidy refers to which of the following phenomenon?

Select one:

- Fusion of male and female gametes producing 2n chromosome numbers in an embryo
- An inversion mutation that includes centromere of chromosomes
- Clubbing together of mitochondria and other cell organelles during interphase
- Presence of one or more extra sets of chromosomes
- Reduplicated nuclear material before cell division

Your answer is incorrect.

Polyploidy refers to the presence of extra sets of chromosomes. Triploidy refers to 3n numbers; tetraploidy refers to 4n numbers. Fusion of male and female gametes producing 2n chromosome numbers in an embryo is the normal diploid state seen in human cells.

The correct answer is: Presence of one or more extra sets of chromosomes

Question 112

Not answered

Marked out of 1.00

Recombination occurs during which phase of meiosis?

Select one:

- Prophase 1
- Anaphase 1
- Interphase
- Telophase 1
- Metaphase 1

Your answer is incorrect.

Chromosomal crossover is an exchange of genetic material between homologous chromosomes. It is one of the final phases of genetic recombination, which occurs during prophase 1 of meiosis in a process called as synapsis. Synapsis is not completed until near the end of prophase I. Crossover usually occurs when matching regions on matching chromosomes break and then reconnect to the other chromosome.

The correct answer is: Prophase 1

Question 113

Not answered

Marked out of 1.00

Ribosome is the site of

Select one:

- Posttranslational modification of protein
- DNA storage
- Protein synthesis
- Synthesis of DNA
- Energy production via respiratory cycle

Your answer is incorrect.

Translation is carried out by more than one ribosome simultaneously.

The correct answer is: Protein synthesis

Question 114

Not answered

Marked out of 1.00

A 53-year-old gentleman presents to the memory clinic with history of progressive short-term memory loss, which was gradual in onset. His MMSE score was 17/30. The gene, which is most likely involved in this condition, is

Select one:

- ApoE4
- Neuronal sortilin related receptor SORL1
- PSEN-1
- Dysbindin
- DISC-1

Your answer is incorrect.

The diagnosis is Alzheimer's disease (early-onset). AD is caused by mutations in the Presenilin-1 (aka PSEN) gene. Mutations in the PSEN1 gene are the most common cause of early-onset AD, accounting for up to 70% of cases

The correct answer is: PSEN-1

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